

Kushwanth Kumar Karamsetti

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Professional Summary

Dynamic and results-driven Data Scientist and Generative AI expert with more than three years of professional experience in designing, deploying, and optimizing AI and data-driven solutions. Proficient in developing scalable machine learning pipelines, fine-tuning large language models, and implementing full-stack web applications. Skilled in leveraging cloud platforms, containerization, and modern frameworks (PyTorch, TensorFlow, LangChain). Adept at collaborating with cross-functional teams to deliver innovative solutions that enhance operational efficiency and drive business growth.

Skills

Programming Languages: Python, R, SQL, Java, PySpark, Linux

Cloud Platforms and DevOps: AWS, Azure, Docker, Kubernetes, Datadog, Amazon Bedrock, Git, GitHub, SSH, Harness, Jenkins, Snowflake, Apache Airflow, ADF, AWS SageMaker, Databricks, Oracle SQL, MongoDB Atlas, PostgreSQL, SQL Server, Kafka, Pandas, PowerBI

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Regression, Hypothesis Testing, Feature Engineering, Model Deployment, Workflow Automation, Model Monitoring, Model Drift Detection, Data Drift Detection

Generative AI: OpenCV, Deep Learning, NLP, Fine-tuning LLMs, Medical Imaging (Segmentation and Classification), Generative AI, RAG Implementation, Re-ranking, Hugging Face, Vector Databases, Embedding Models, Prompting Techniques, Whisper AI, Azure AI Studio, Langchain, LangGraph, LlamaIndex, Haystack, Vertex AI

Software Frameworks and Tools: Flask, FastAPI, SciPy, Redis, gRPC, NumPy, Pandas, Agile (Scrum), CI/CD, MLOps/LLMOps (Airflow, Mlflow), Data Orchestration, WhyLabs, Splunk, Azure Cosmos DB, PowerBI

Work Experience

Senior Data Scientist at Tror - AI for Everyone

Duration: Dec 2022 – Present **Location:** Nashville, TN

- Developed a Generative AI chat agent using Langchain, LangGraph, LlamaIndex, and Haystack, incorporating multi-agent RAG workflows for seamless user-specific interactions.
- Integrated multi-agent AI chatbots with RAG-enabled knowledge bases for document support, web search, tool calls, and secure fine-tuned model usage; leveraged MongoDB for chat history storage.
- Streamlined workflows with Apache Airflow to process documents and embed data using open-source models.
- Built APIs with FastAPI and gRPC microservices; integrated Confluence and GDrive for knowledge bases.
- Improved chat performance with Redis Cache, minimizing database queries and reducing costs.
- Adopted Agile Scrum practices with Jira for sprint planning and team collaboration.
- Utilized APIs (GPT, Claude) and deployed open-source models like LLaMA on AWS SageMaker with Amazon Bedrock integration.
- Integrated Stripe for payment processing and implemented voice transcription to enhance accessibility.
- Created medical image segmentation models for brain and pancreas tumor detection using PyTorch.
- Developed a voice transcription app with 95.6% accuracy for North American accents using Whisper AI.
- Utilized Kafka to build real-time data pipelines for efficient data streaming.

- Monitored ML models for drift and performance issues using WhyLabs, Splunk, and Datadog.
- Managed the full lifecycle of ML models, including deployment and integration into data pipelines and BI tools.
- Enhanced data reliability using Azure Cosmos DB, AKS, Databricks, and MLOps best practices.
- Implemented Docker and Kubernetes for containerization and orchestration of AI workflows.
- Optimized PyTorch and TensorFlow pipelines on Ubuntu for resource-efficient training and inference.
- Streamlined CI/CD pipelines for AI applications with Harness and Jenkins.

Data Engineer at Client: Kimberly-Clark, Tredence Solutions Private Limited

Duration: May 2022 – Dec 2022 **Location:** Bangalore, India

- Developed and managed data pipelines with Azure Data Factory (ADF), Snowflake, and Databricks to optimize data transformations.
- Utilized PySpark on Azure Databricks for processing and analyzing large datasets.
- Integrated Oracle SQL for advanced querying and data manipulation.
- Implemented Apache Airflow for task scheduling, monitoring, and error handling.
- Employed Azure DevOps for version control, CI/CD pipelines, and automated deployments.
- Used Azure Boards for sprint planning and Agile task tracking.
- Applied SDLC practices for scalable data engineering solutions.

Data Scientist Intern at Tror - AI for Everyone

Duration: May 2021 – May 2022 **Location:** Hyderabad, India

- Contributed to a Master Data Management (MDM) backend workflow using Langchain and PyTorch, integrating Redis cache for efficient data retrieval.
- Assisted in a lung cancer detection project by implementing preprocessing techniques with OpenCV and TensorFlow, and improving segmentation with transformer-based models.
- Developed an ETL pipeline with Apache Airflow to preprocess loan application data and deploy a binary classification ML model.

Education

Masters in Information Technology Management at Webster University

Duration: Sep 2023 – Dec 2024 **Location:** San Antonio, TX
GPA: 3.93/4.0

Bachelor of Technology in Mechanical Engineering at National Institute of Technology, Surat (SVNIT)

Duration: Jul 2018 – May 2022 **Location:** Surat, India
GPA: 8.02/10.0